

Hybrid IAM (Identity & Access Management) Lab

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Table of Contents

Sr no.	Title	Pg no.
1.	Overview	3
2.	Infrastructure & Architecture	3
3.	Active Directory & RBAC	4
4.	Hybrid Identity & Cloud Sync	7
5.	Authentication & Application Access	9
6.	Identity Lifecycle Management (JML)	10
7.	Monitoring, Audit & Troubleshooting	11
8.	Firewall	11
9.	Conclusion	13

1. Overview

Built a hybrid identity and access management environment integrating on-premises Active Directory with Microsoft Entra ID. The lab simulates common IAM operations including identity provisioning, group-based access control, MFA, application access, joiner-mover-leaver workflows, and IAM troubleshooting.

Objective

To gain hands-on experience administering identities and access across an on-premises Active Directory and cloud-based Microsoft Entra environment.

Technologies

Windows Server 2025, Active Directory Domain Services, Microsoft Entra ID, Entra Cloud Sync, Windows 10, pfSense, VMware Workstation

2. Infrastructure & Architecture

The lab uses a hybrid infrastructure consisting of an on-premises Windows Server 2025 domain environment, a Windows client, pfSense for network security, and Microsoft Entra ID for cloud-based identity and access management.

DC01 - Windows Server 2025: Active Directory Domain Services

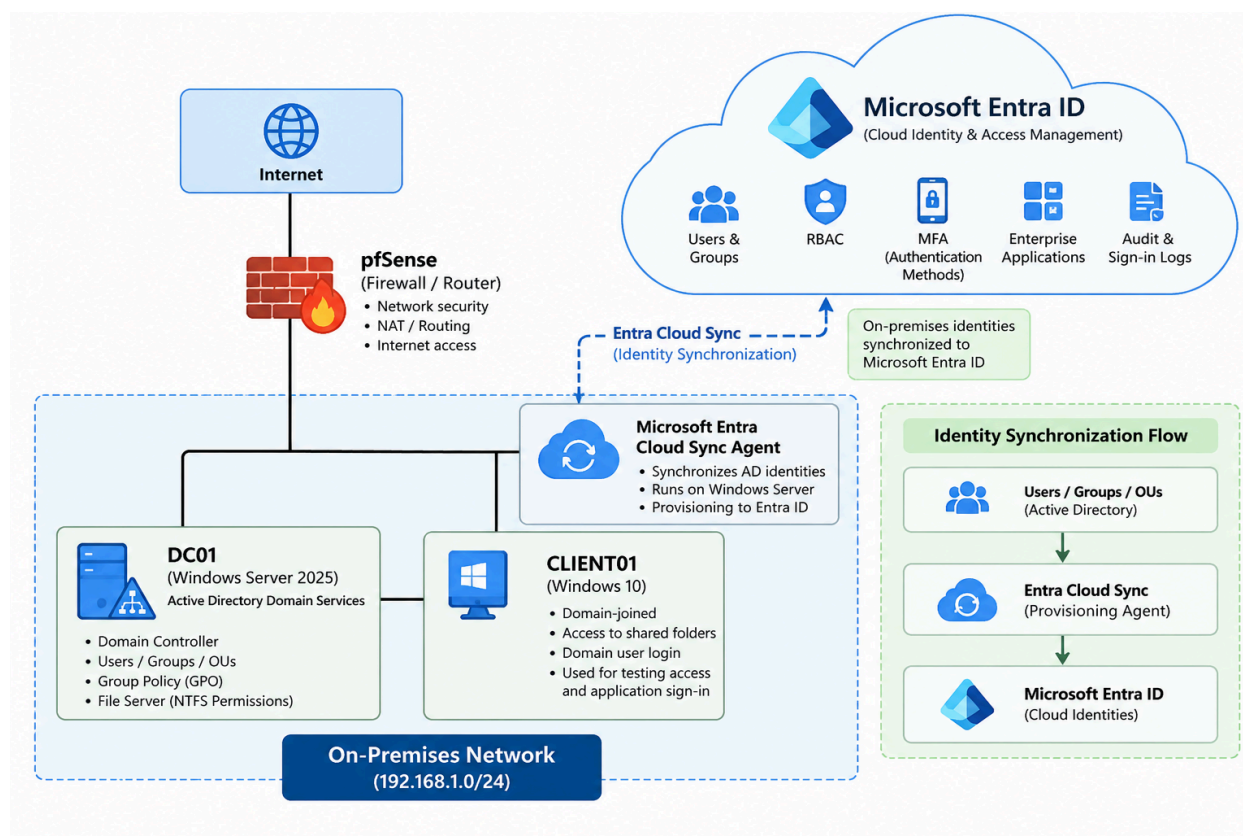
CLIENT01 - Windows 10: Domain-joined client

pfSense: Firewall, routing and network services

Microsoft Entra ID: Cloud identity and access management

Entra Cloud Sync: Synchronizes on-premises identities to Entra ID

Architecture diagram



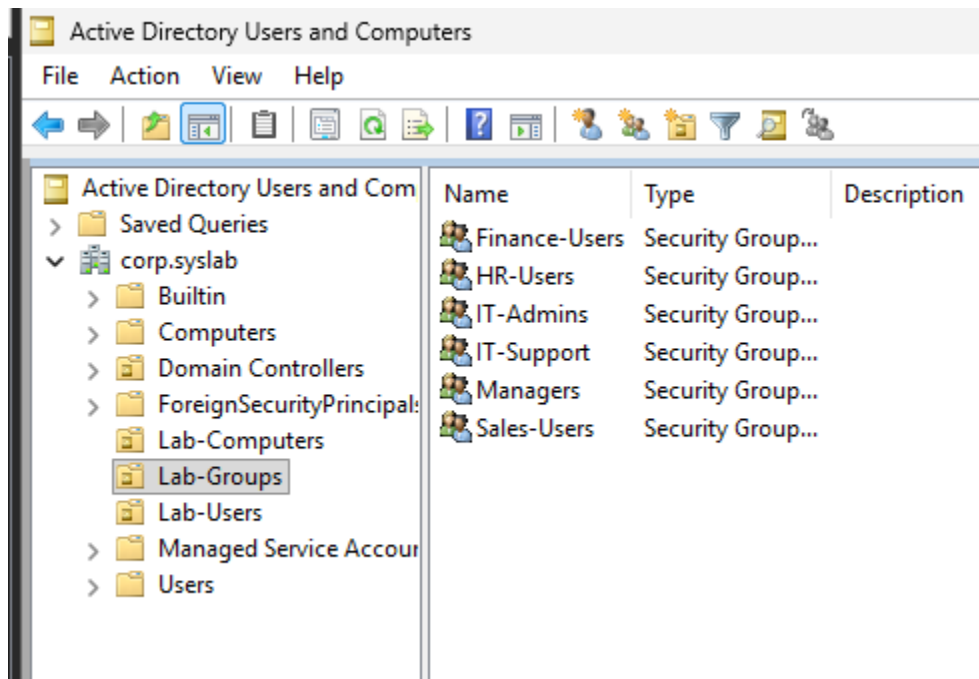
3. Active Directory & Role-Based Access Control

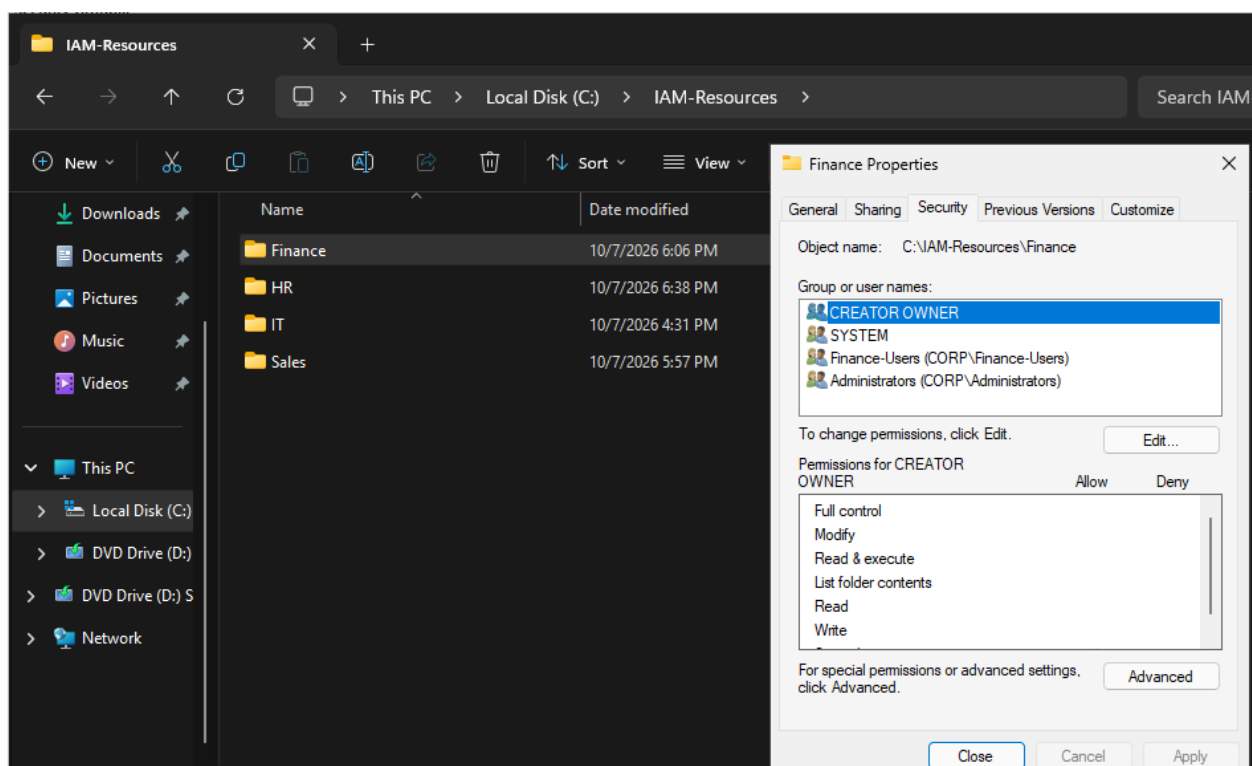
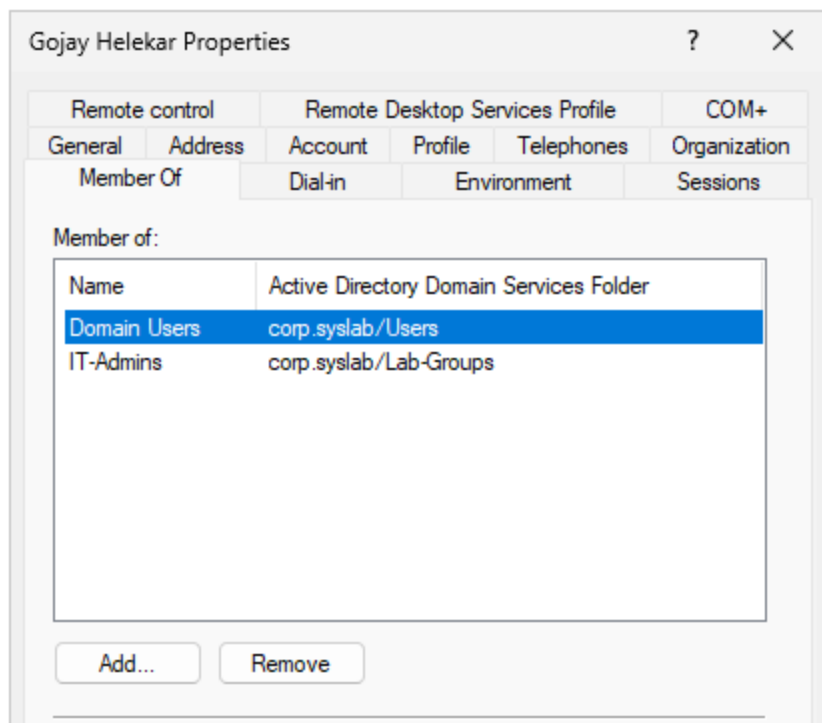
Configured Active Directory as the on-premises identity source, creating organizational units, users, and department-based security groups. Implemented group-based access control and NTFS permissions to restrict resources according to departmental roles.

Implemented

- Users & OUs - Organized identities into structured OUs.
- Department Groups - Finance, HR, IT, Sales, Managers, and IT-Support.

- RBAC - Assigned users to appropriate security groups based on their role.
- NTFS Permissions - Applied department-specific access to shared folders.
- Group Policy - Configured domain-level policies for managed client systems.





4. Hybrid Identity & Cloud Sync

Integrated the on-premises Active Directory environment with Microsoft Entra ID using Microsoft Entra Cloud Sync. User identities created and managed in AD were synchronized to the cloud directory, enabling centralized cloud identity and access management.

Implemented

- Configured Microsoft Entra Cloud Sync
- Installed and configured the Provisioning Agent on DC01
- Established AD → Entra ID synchronization
- Verified synchronized users in Entra ID
- Used provisioning logs to monitor synchronization
- Tested provision-on-demand
- Investigated and resolved a provisioning quarantine issue

+ New configuration ▾ ↻ Refresh | 🗉 Got feedback?

Sync identities from on-premises Active Directory to Microsoft Entra ID. Read the [configuration guide](#) [↗](#) for help configuring sync.

Configuration	Sync direction	Status	Actions
corp.syslab	AD to Microsoft Entra ID	● Healthy	...

↓ Download on-premises agent

All agents registered to this tenant

Machine Name	External IP	status
DC01.corp.syslab	152.57.123.140	✓ active

Date	Identity	Action	Source system
10/7/26, 9:39:17 PM	Display name Ranjan Gowda Source ID 62d3ca47-727a-4fff-8413-e2d6428c0d06 Target ID dc7630f9-a895-4693-8335-67efc3b13eef	Create	Active Directory
10/7/26, 9:53:39 PM	Display name Ranjan Gowda Source ID 62d3ca47-727a-4fff-8413-e2d6428c0d06 Target ID dc7630f9-a895-4693-8335-67efc3b13eef	Update	Active Directory
10/8/26, 6:13:48 PM	Display name Ranjan Gowda Source ID 62d3ca47-727a-4fff-8413-e2d6428c0d06 Target ID dc7630f9-a895-4693-8335-67efc3b13eef	Update	Active Directory

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5. Authentication & Application Access

Configured Microsoft Entra authentication methods and practiced MFA enrollment while managing access to an Enterprise Application through Entra ID.

Implemented

- Configured Microsoft Authenticator as an authentication method.
- Created an MFA registration campaign for a test user.
- Configured an Enterprise Application in Microsoft Entra ID.
- Assigned a user to the application.
- Practiced Entra-based application access management.

Start a registration campaign that prompts users to set up more secure authentication methods., currently limited
[Learn more](#) 

State	Enabled
Authentication method	Microsoft Authenticator
Days allowed to snooze	1
Limited number of snoozes	Disabled
Include	Select users or groups 1 User Select group
Exclude	None selected Select group

Method	Target	Enabled
▼ Built-In		
Passkey (FIDO2)	All users	Yes
Microsoft Authenticator	All users	Yes
SMS		No

Home > Microsoft Entra Connect | Cloud Sync > Cloud sync | Provisioning logs > Authentication methods | Policies > Enterprise applications | All applications > IAM-SAML-Test-App

IAM-SAML-Test-App | Users and groups

Enterprise Application

Overview

Deployment Plan

Diagnose and solve problems

Manage

Properties

Owners

Roles and administrators

Users and groups

Single sign-on

«

+ Add user/group

✎ Edit assignment

🗑 Remove assignment

🔍 Update credential

🔄 Refresh

⚙ Manage view

🗨 Got feedback?

🔔 The application will appear for assigned users within My Apps. Set "visible to users?" to no in properties to prevent this.

Assign users and groups to app-roles for your application here. To create new app-roles for this application, use the [application registration](#)

🔍 First 200 shown, search all users & groups

Display name	Object type
☐  Ranjan Gowda	User

6. Identity Lifecycle (JML)

Demonstrated identity lifecycle management by provisioning users through Active Directory, synchronizing identities to Microsoft Entra ID, and updating user attributes and group memberships as roles changed.

Ranjan Gowda | Groups

Search

«

+ Add memberships

✕ Remove memberships

🔄 Refresh

☰ Columns

🗨 Got feedback?

🔍 Search groups

+ Add filters

	Name	Object Id	Group Type	Membership Type	Email	Source
☐	 IT-Support	6d0ce5a2-db5b-4daf-9...	Security	Assigned		Windows Server AD

Overview

Audit logs

Sign-in logs

Diagnose and solve problems

Custom security attributes

7. Monitoring, Audit & Troubleshooting

Used Microsoft Entra provisioning and audit logs to monitor identity synchronization, track user changes, and troubleshoot synchronization issues within the hybrid identity environment.

Implemented

- Monitored Cloud Sync provisioning logs
- Verified successful Create / Update operations
- Reviewed Entra audit logs for identity changes
- Investigated a provisioning quarantine condition
- Verified the provisioning agent and AD availability to restore healthy synchronization

Date	↑↓	Service	Category	↑↓	Activity	↑↓	Status	Status reason	Target(s)
10/8/2026, 6:39:20 PM		Core Directory	UserManagement		Add app role assign...		Success		IAM-SAML-Test-A
10/8/2026, 6:13:48 PM		Core Directory	UserManagement		Update user		Success		ranjan@helekarsc
10/7/2026, 9:53:39 PM		Core Directory	UserManagement		Update user		Success		ranjan@helekarsc
10/7/2026, 9:53:37 PM		Core Directory	UserManagement		Update StsRefreshTo...		Success		ranjan@helekarsc
10/7/2026, 9:53:37 PM		Core Directory	UserManagement		Change user password		Success		ranjan@helekarsc
10/7/2026, 9:53:37 PM		Core Directory	UserManagement		Update PasswordPro...		Success		ranjan@helekarsc
10/7/2026, 9:39:17 PM		Core Directory	UserManagement		Add user		Success		ranjan@helekarsc
10/7/2026, 9:39:17 PM		Core Directory	UserManagement		Update PasswordPro...		Success		ranjan@helekarsc

8. Network Security & Firewall

Deployed pfSense as the network gateway for the lab environment, providing firewall and DNS services for the infrastructure. Verified network connectivity between the lab systems through the pfSense-managed network.

Implemented

- Deployed pfSense as the lab firewall.
- Configured the lab network through pfSense.
- Used pfSense for DNS and network connectivity.
- Verified connectivity between the Windows Server, Windows client, and Linux server.
- Used the firewall as the network boundary for the lab environment.

```
Starting CRON... done.
Starting package System Patches...done.
pfSense 2.9.0-RELEASE amd64 20260918-1643
Bootup complete

FreeBSD/amd64 (pfSense.home.arpa) (ttyv0)

VMware Virtual Machine - Netgate Device ID: 9bb9f7bd73ee312ea12e

*** Welcome to pfSense 2.9.0-RELEASE (amd64) on pfSense ***

WAN (wan) -> em0 -> v4/DHCP4: 192.168.162.154/24
LAN (lan) -> em1 -> v4: 192.168.10.1/24

0) Logout / Disconnect SSH          9) pfTop
1) Assign Interfaces                10) Filter Logs
2) Set interface(s) IP address     11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults       13) Update from console
5) Reboot system                  14) Enable Secure Shell (sshd)
6) Halt system                    15) Restore recent configuration
7) Ping host                      16) Restart PHP-FPM
8) Shell

Enter an option: █
```

9. Conclusion

This project provided hands-on experience building and operating a hybrid Identity and Access Management environment using on-premises Active Directory and Microsoft Entra ID. The lab covered identity provisioning, directory synchronization, RBAC, NTFS access control, MFA, application access, identity lifecycle workflows, audit monitoring, and IAM troubleshooting, with pfSense supporting the underlying network infrastructure.

The project also provided practical exposure to real-world IAM constraints, including Microsoft Entra licensing limitations, where certain capabilities such as group-based Enterprise Application assignment and Access Reviews require higher-tier licensing. These limitations were documented and considered as part of the IAM administration and troubleshooting process.